

Alejandro Jaimes

alejaimes.dev | ale@alejaimes.dev | linkedin.com/in/alejaimes | github.com/alecocosette

EDUCATION

University of Central Florida

Bachelor of Science in Computer Science; Minor in Robotics

Orlando, FL

May 2028

EXPERIENCE

Software Engineer Intern

May 2026 - Present

Institute for Simulation and Training

Orlando, FL

- Architected shared spatial-anchor system with **Unity** using **AR Foundation**, aligning **+10** users within **5cm**.
- Developed a periodic re-localization system correcting SLAM-induced spatial drift every **2** seconds, maintaining anchor accuracy for users navigating **50+** meters from the origin point.
- Designed backward-compatible multiplayer architecture across **2** codepaths, preserving **100%** single-player.
- Streamed anchor-relative transforms via **Photon RPC** at **10Hz** for real-time multi-user tracking.

President

Apr 2026 - Present

Graphics Programming Knights

Orlando, FL

- Established Graphics Programming Knights, a non-profit student organization, growing membership to **165+**.
- Organized the first community-wide Render Jam competition resulting in **40+** participants and **5** projects.
- Orchestrated **2 workshops per term** on pathways into graphics programming, covering essential **mathematics**, **game development engines**, and **GPU programming** with student attendance of **20+**.
- Expanded club reach by **40%**, establishing partnerships to have workshops and hackathons with **3** organizations.

Outreach Board

Jan 2026 - Present

Knight Hacks

Orlando, FL

- Boosted online community engagement online with strategic content creation and event awareness by **57%**.
- Produced and edited Reels, increasing Knight Hacks' visibility and follower engagement, averaging **5k+** views.
- Expanded event attendance by **15%** compared to previous semesters through targeted outreach campaigns on social media and speaking to the public, increasing participation with an average of **40+** people per meeting.

PROJECTS

greghouse | *C++, Arduino, ESP32, React, AWS, Node.js, Lambda, DynamoDB*

- **3rd place winner** of the **Best Overall** category at BloomKnights out of **86** projects and **240+** participants.
- Engineered **ESP32** firmware in **C++** streaming soil telemetry to **AWS** every 2s with sub-3s latency.
- Architected an IoT pipeline with **API Gateway**, **Lambda**, and **DynamoDB** handling **1,800+** readings hourly.
- Crafted a pixel-art **TFT** display on **C++** with mood-driven animations, synchronizing with the **React** app.

sFun (SNES Emulator) | *C++, CMake, JSON, SDL2*

- Architected a **65C816 CPU** emulator in **C++** with 8/16-bit width switching, and 24-bit addressing.
- Built a **256-entry function pointer** dispatch table decomposed in **9** logical translation units for maintainability.
- Implemented **50+ CPU instructions** covering arithmetic, logic, branching, stack, and subroutine calls with correct flag behavior.
- Engineered a **JSON-driven harness** with register-level diffing, achieving **100%** pass rate across **25+** test cases.

Next Step | *React, Tailwind, Next.js, TypeScript, Python, Flask, Vapi*

- **3rd place winner** of State Farm's Sponsor Challenge at Shell Hacks 2025 out of **50** projects submitted.
- Created an AI voice agent delivering interactive, personalized feedback across **3** different insurance categories.
- Implemented backend services to process and structure conversational data generating dynamic agent responses, reducing response time by **66%** using **Python**, integrated with the **Vapi API**.
- Engineered **7** interactive components for optimal experience with **TypeScript**, **Next.js**, and **React** for frontend.

TECHNICAL SKILLS

Languages: C, C++, C#, Python, HTML, CSS, JavaScript, TypeScript, Assembly, Java

Frameworks: Unity, Next.js, Node.js, Flask, Unreal Engine, Three.js, OpenCV, Mediapipe, CMake, OpenGL, CUDA

Libraries: Tailwind CSS, React, SDL2, WebSockets, TFT_eSPI, Arduino

Tools: ESP32, Docker, AWS, Cloudflare, Git, Arduino IDE, PlatformIO, Vite